

Homework

MIT 6.792 Reinforcement Learning

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Collaborators: Claude 4.5 Sonnet (Questions, not Solving) · Resources Used: None

$$\nabla_{\theta} V_{\mu}(\pi_{\theta}) = \mathbb{E}_{\tau \sim P} \left[R(\tau) \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] \quad (1)$$

$$= \mathbb{E}_{\tau \sim P} \left[\left(\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right) \left(\sum_{t'=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_{t'} | s_{t'}) \right) \right] \quad (2)$$

$$= \mathbb{E}_{\tau \sim P} \left[\sum_{t=0}^{\infty} \sum_{t'=0}^{\infty} \gamma^t r(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_{t'} | s_{t'}) \right] \quad (3)$$

$$= \mathbb{E}_{\tau \sim P} \left[\sum_{t'=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_{t'} | s_{t'}) \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad (4)$$

$$= \mathbb{E}_{\tau \sim P} \left[\sum_{t'=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_{t'} | s_{t'}) \left(\sum_{t=0}^{t'-1} \gamma^t r(s_t, a_t) + \sum_{t=t'}^{\infty} \gamma^t r(s_t, a_t) \right) \right]$$

- List test
- List test
- List test
 - List test

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turba omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeo, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo

docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus.

<https://colab.research.google.com/drive/11VOd7wX0xWXR4uUfNZWP8fDIygm-QmR?usp=sharing>

$a \cdot b \cdot c^* @ v \approx u o . v \cdot \text{Pr}[X] \cdot \text{Var}[Y]$ test

1 Other packages of interest:

1.1 Scribe (High likelihood)

Uses (mostly) ascii math to simplify the look of operators

With scribe you can write $f \circ (f \circ g)(x) \setminus \{0\} \mapsto \pm 1 \pmod n$

instead of $\text{cal}(F) \text{ dot.op } (f \text{ compose } g)(x) \text{ without } \{0\} \text{ equiv plus.minus } 1 \text{ quad } (\text{mod } n)$

1.2 Math Shorthands

Interesting process of looking for and replacing elements inside of your body. Could probably steal from this. See: <https://github.com/EpicEricEE/typst-quick-maths/blob/master/src/quick-maths.typ#L46>

1.3 Eqalc: Convert Math equations to functions

<https://typst.app/universe/package/eqalc>

Along a similar vein, numty (like numpy): <https://typst.app/universe/package/numty>

Fitch-style proofs?? Prob not but interesting. <https://typst.app/universe/package/derive-it>

1.4 Prismath

Seems cool but also doesn't seem to work well lol. May need to add with show to be better. <https://typst.app/universe/package/prismath>

2 Default Typst Stuff

Defining a Math Operator

<https://typst.app/docs/reference/math/class/>

$\cong$

For math symbols (note: single letters cannot be overridden)

$\forall$

$\sum$

$x \heartsuit y \wedge y \heartsuit 5$

For macros:

P

$\text{func}_X[X^2]$

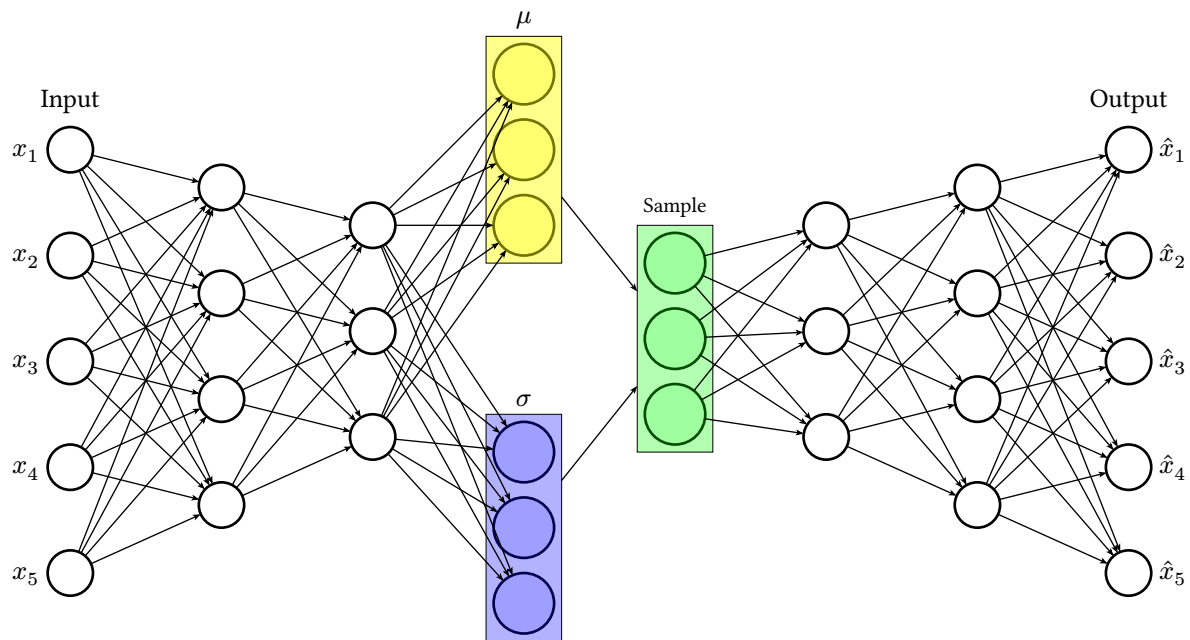
3 My Macros

Question
Done
Partially complete
todo
Hey team
Hey team
Hey team
Hey team

4 Diagramming

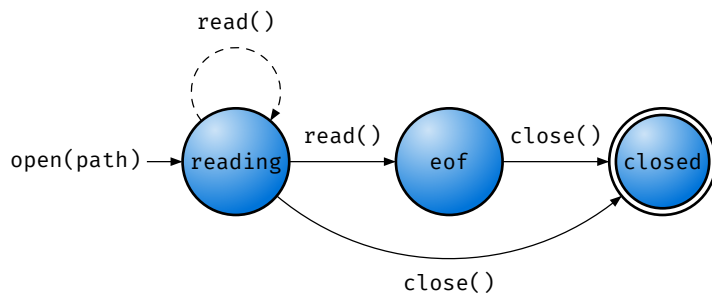
4.1 CeTZ (Drawing)

- [Manual](#)
- [My Personal Notes](#)

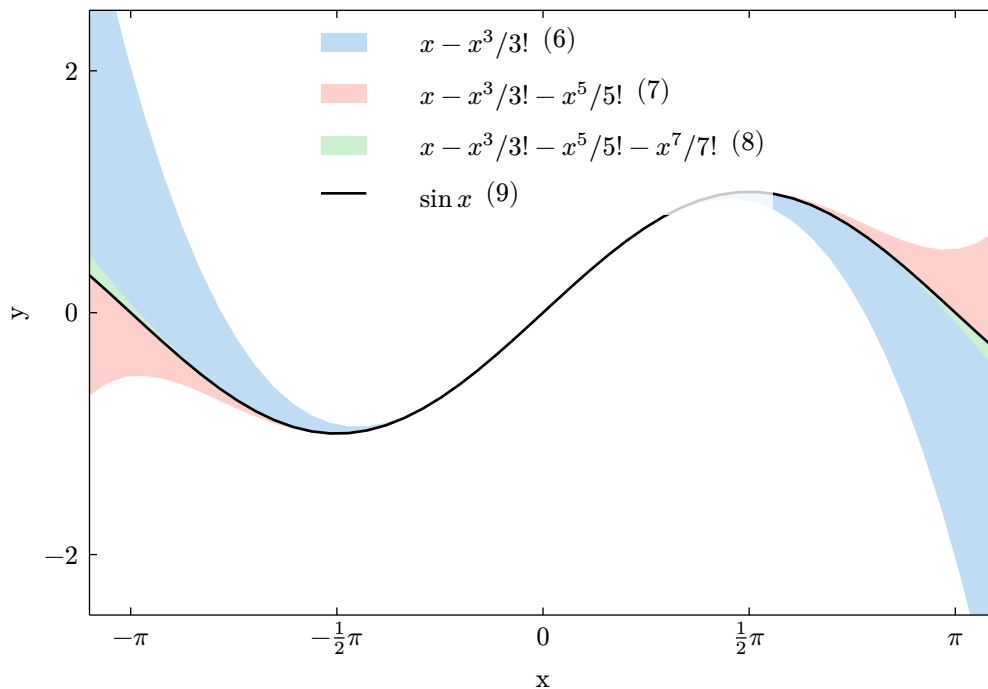


4.2 Fletcher (Arrow Diagrams)

- [Manual](#)

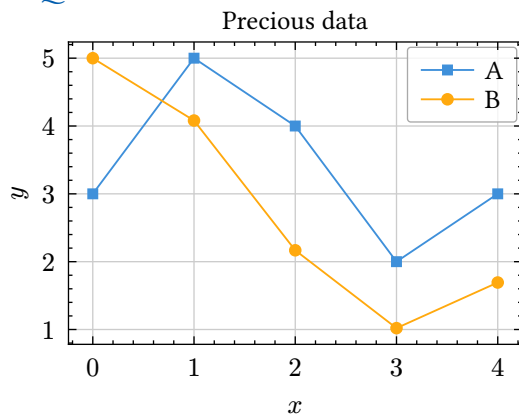


4.3 CeTZ Plot (Plotting)



4.4 Lilaq (Data Visualization, Slightly better than CeTZ Plot)

- Very well-maintained
- [Quickstart](#)



5 Math

5.1 MiTeX

$$z = \begin{bmatrix} x_{0:T} \\ u_{0:T-1} \\ \lambda_{0:T-1} \end{bmatrix} \quad (10)$$

5.2 Pavemat

- Not imported since rare. Can define paths within matrix cells or fills of cells.

5.3 Great Theorems

- Again not imported, used for keeping track of various theorems

5.4 mannot

$$\underbrace{p_i}_{\text{Probability of state } i} = \frac{\underbrace{\exp(-\beta E_i)}_{\substack{\text{Boltzmann factor} \\ \text{Energy}}}}{\underbrace{\sum_j \exp(-\beta E_j)}_{\text{Partition function}}} \quad (11)$$

6 Display

6.1 frame-it (Custom Frames)

- Don't fully understand how these are used
- How to use it is explained below. Here is a quick example:

TitleOptional Tag

Body, i.e. large content block for the frame.

Example 1

Definition 1:

TitleOptional Tag

Body, i.e. large content block for the frame.

Example 2

TitleOptional Tag

Body, i.e. large content block for the frame.

Feature 3

- Variant 4 **Title** (*Optional Tag*): Body, i.e. large content block for the frame.
- Syntax 5 **Title** (*Optional Tag*): Body, i.e. large content block for the frame.

6.2 Hydra

- Query and display headings in your documents
- Single function

6.3 Physica

- Inner product iprod - $\langle u,v\rangle$
- Vector Bold vb - $\boldsymbol{v}$
- Vector Unit vu - $\hat{\boldsymbol{u}}$
- Diagonal Matrix dmat - $\begin{pmatrix} 1 & \\ & \mathbb{R} \end{pmatrix}$ (mdet, zmat, imat, jmat, rot2mat, rot3{xyz}mat,) = (determinant, zero mat, identity, jacobian)
- Differential dd - $\mathrm{d}x$
- Hessian hmat(function; var1, var2) - $\begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$
- Set Set(item, condition) - $\{x_i \mid i \in \{1:3\}\}$
- My template has superscript T become transpose
- See manual <https://github.com/Leedehai/typst-physics/blob/master/physica-manual.pdf>

$A^\top, \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \quad \Lambda^\mu{}_\nu = \begin{pmatrix} 1 & \\ & \mathbb{R} \end{pmatrix}, \quad f(x,y) \, \mathrm{d}x \, \mathrm{d}y, \quad \mathrm{d}^3 \boldsymbol{x} \, \mathrm{d}y, \quad \Delta^2 x \wedge \Delta^2 y, \quad \frac{\mathrm{D} \varphi}{\mathrm{D} t} = \frac{\partial \varphi}{\partial t} + \boldsymbol{u} \nabla \varphi$

(12)

$H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}, \quad \boldsymbol{v}^a = \sum_{i=1}^n \alpha_i \hat{\boldsymbol{u}}^i, \quad \left\{ (x,y) \left| \frac{\partial^3 f}{\partial x^2 \partial y} + \frac{\partial^3 f}{\partial x \partial y^2} < \varepsilon \right. \right\}$

(13)

$-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} \psi + \nabla^2 \psi = \frac{m^2 c^2}{\hbar^2} \psi, \quad |n^{(1)}\rangle = \sum_{k \notin D} \frac{\langle k^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle, \quad \int_V \mathrm{d}V \left(\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \right) \right) = 0$

(14)

$\mathrm{d}^2 s = - \left(1 - \frac{2GM}{r} \right) \mathrm{d}^2 t + \left(1 - \frac{2GM}{r} \right)^{-1} \mathrm{d}^2 r + r^2 \mathrm{d}^2 \Omega$

(15)

$\text{clk: } \text{[square wave]}$

(16)

$\text{bus: } \text{[bus signal]}$

(17)

7 Code

7.1 Codly

Should just work by default, is initialized at the start to style code blocks

```
1 pub fn main() {  
2     println!("Hello, world!");  
3 }
```

 Rust

7.2 Callisto

- In = Code or markdown of cell
- Out = Output of a cell (Can't handle HTML)
- Cell = In + Out
- Render = Not sure how this differs from Cell

All of these can take a few arguments

- Main positional - Index, list of indices, OR a string representing the tag of the cell(s).
 - Tag is probably the best usage.

```
1 x = 5
2 y = 10
3 x + y
```

Python

In []:

```
1 from plotly.subplots import make_subplots
2
3
4 def plot_optimal_simulation(
5     T: int,
6     A: np.ndarray,
7     B: np.ndarray,
8     Q: np.ndarray,
9     Q_f: np.ndarray,
10    r: float,
11    x_0: np.ndarray,
12    dynamics: Callable,
13    title: str | None = None,
14 ) → tuple[list[np.ndarray | None], list[np.ndarray | None]]:
15     x_optimal, u_optimal = simulate_optimal(T, A, B, Q, Q_f, r, x_0, dynamics)
16     p_optimal = [x[0][0] for x in x_optimal]
17     v_optimal = [x[1][0] for x in x_optimal]
18
19     title = f"[Optimal Simulation] Q = {Q}, r = {r}, Q_f = {Q_f}" if title is None else title
20
21     # To match the time steps, x is one longer
22     # u_optimal.append(np.ndarray([0]))
23
24     t = list(range(len(p_optimal)))
25     fig = make_subplots(specs=[[{"secondary_y": True}]]))
26     # Optimal Position Trace
27     fig.add_trace(
28         go.Scatter(
29             name="Optimal Position (x_t)",
30             x=t,
31             y=p_optimal,
32         )
33     )
34     # Optimal Acceleration Trace
35     u_optimal = [u[0][0] for u in u_optimal]
36     fig.add_trace(go.Scatter(x=t, y=u_optimal, name="Optimal Acceleration (u_t)",
37                             secondary_y=True))
37     fig.update_layout(title=title, xaxis_title="Time Step, t", yaxis_title="Position")
```

Python

7.2 Callisto

```
38     fig.update_yaxes(title_text="Acceleration", secondary_y=True)
39
40     return fig
41
42
43 for penalty_name, param_set in penalty.items():
44     for Q, r, Q_f in param_set:
45         fig = plot_optimal_simulation(T, A, B, Q, Q_f, r, x_0, dynamics)
46         fig.show()
```